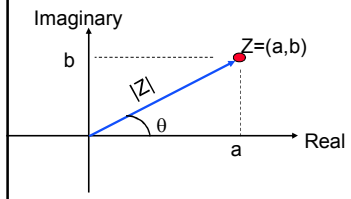
Complex Numbers



$$Z = \mathrm{Re}(Z) + i\,\mathrm{Im}(Z) = a + ib = |Z|e^{i\theta}$$

$i = \sqrt{-1} \quad (i^2 = -1)$	
$e^{i\theta} = \cos\theta + i\sin\theta$	(a unit vector)
$\lvert Z\rvert = \sqrt{a^2 + b^2}$	(Magnitude)
$\theta = tg^{-1}(b/a)$	(phase)
$\bar{Z} = Z^* = a - ib = \lvert Z\rvert e^{-i\theta}$	(conjugate)

The Continuous Fourier Transform

1D Continuous Fourier Transform:

$$f(x) = \int_{u=-\infty}^{\infty} \mathrm{F(u)} e^{+2\pi i u x} du = \Im^{-1}(F(u))$$ The Inverse Fourier Transform

$$F(u) = \int_{x=-\infty}^{\infty} \mathrm{f(x)} e^{-2\pi i u x} dx = \Im(f(x))$$ The Fourier Transform

2D Continuous Fourier Transform:

$$f(x,y) = \int_{u=-\infty}^{\infty}\int_{v=-\infty}^{\infty} \mathrm{F(u,v)} e^{+2\pi i(ux+vy)} du dv$$ The Inverse Transform

$$F(u,v) = \int_{x=-\infty}^{\infty}\int_{y=-\infty}^{\infty} \mathrm{f(x,y)} e^{-2\pi i(ux+vy)} dx dy$$ The Transform

The Discrete Fourier Transform

1D Discrete Fourier Transform:

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x) e^{\frac{-2\pi i u x}{N}}$$ (u = 0,..., N-1)

$$f(x) = \sum_{u=0}^{N-1} F(u) e^{\frac{+2\pi i u x}{N}}$$ (x = 0,..., N-1)

2D Discrete Fourier Transform:

$$F(u,v) = \frac{1}{N}\frac{1}{M}\sum_{x=0}^{N-1}\sum_{y=0}^{M-1} f(x,y) e^{-2\pi i\left(\frac{ux}{N}+\frac{vy}{M}\right)}$$ (u = 0,..., N-1; v = 0,...,M-1)

$$f(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{M-1} F(u,v) e^{+2\pi i\left(\frac{ux}{N}+\frac{vy}{M}\right)}$$ (x = 0,..., N-1; y = 0,...,M-1)

Some Fourier Transforms

Function	Fourier Transform
$f(x) = \mathrm{rect}(x) = \begin{cases} k & -c \le x \le c \\ 0 & otherwise \end{cases}$	$F(u) = 2kc \cdot \mathrm{sinc}(2\pi c u)$
$f(x) = \delta(x)$	$F(u) \equiv 1$
$f(x) = \delta(x-\alpha)$	$F(u) = e^{-2\pi i u \alpha}$
$f(x) = \cos(\alpha x)$	$F(u) = \frac{1}{2}\delta\left(u - \frac{\alpha}{2\pi}\right) + \frac{1}{2}\delta\left(u + \frac{\alpha}{2\pi}\right)$
$f(x) = \sin(\alpha x)$	$F(u) = \frac{1}{2i}\delta\left(u - \frac{\alpha}{2\pi}\right) - \frac{1}{2i}\delta\left(u + \frac{\alpha}{2\pi}\right)$
$\mathrm{Gaussian}(x) = e^{-x^2/b}$	$\mathrm{Gaussian}(u) = \sqrt{\pi b} \cdot e^{-\pi^2 u^2 b}$

About the Discrete Transform

1. f_d is a discrete sampling of $f(x)$ at gaps Δx

   ⇕

   F_d is a discrete sampling of $F(u)$ at gaps $\Delta u = \frac{1}{N\Delta x}$

2. **Periodicity of the discrete transform (both $f(x)$ and $F(\omega)$):**
   $F(\omega+N) = F(\omega)$
   $f(x+N) = f(x)$

   $F(\omega+N) = \sum_0^{N-1} f(x)\, e^{-2\pi i x(\omega+N)/N} = \sum_0^{N-1} f(x)\, e^{-2\pi i x\omega/N} \underbrace{e^{-2\pi i x}}_{=1}$
   $f(x+N) = f(x)$ *can be shown using the inverse transform.*

3. **Computational complexity:** $O(N \cdot \log N)$
   (with FFT – the Fast Fourier Transform)

Fourier Properties

1. **Separability of the 2D transform:**
   2D Transform = successive applications of 1D transforms

2. **Linearity:** $f+g \leftrightarrow F+G$
   $af \leftrightarrow aF$

3. **Image average:** $F(0,0) = avg(f(x,y))$

4. **Shift ⇔ phase-change:** $g(x)=f(x+a) \leftrightarrow G(\omega)=F(\omega)e^{2\pi i\omega a}$ *(continuous)*
   $G(\omega)=F(\omega)e^{2\pi i\omega a/N}$ *(discrete)*
   ⇓
   $|G(\omega)| = |F(\omega)|$

   * *Same magnitude; only phase shift.*
   * *Can be used to recover shift (a,b) between two images f(x,y) & g(x,y):*
     $G(u,v)/F(u,v)=e^{2\pi i(\frac{ua}{N}+\frac{vb}{M})} \leftrightarrow \delta(x+a,y+b)$

Fourier Properties

5. **Scaling/Shrinking:** $f(ax) \leftrightarrow \frac{1}{|a|} F\left(\frac{u}{a}\right)$

6. **Derivative:** $g(x)=\frac{d}{dx}f(x) \leftrightarrow G(\omega)= 2\pi i\, \omega\, F(\omega)$
   * *G(0)=0.*
   * *Taking the derivative increases high frequencies (noise, edges)*
     ➔ *like a high-pass filter (HPF).*

7. **Rotation:** *rotation of f(x,y) by θ ↔ rotation of F(u,v) by θ*
   (can be shown using polar coordinates)

8. **Principle of Uncertainty:** No meaning to *a frequency at a point* !